

# **CREDIT RISK ASSESSMENT: PREDICTING CREDIT CARD DEFAULT USING CUSTOMER AND CREDIT CHARACTERISTICS**

**Project Report Submitted in fulfillment of the  
Requirements for the Award of the Internship of  
Summer Training Program by IDRA**

Data Science and AI

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## DECLARATION

I hereby declare that the project report entitled “**Predictive Analytics for Credit Risk Assessment: Machine Learning-Based Prediction of Credit Card Default**” is my original work carried out as part of the requirements of the **Summer Training Program in Data Science and Artificial Intelligence** conducted by the **India Data Research Academy (IDRA)**.

I declare that this project has been prepared based on my study, analysis, implementation, and interpretation of the provided credit card default dataset. The project follows a systematic data science workflow involving problem definition, dataset understanding, data cleaning, preprocessing, exploratory data analysis, statistical analysis, visualization, machine learning model development, model evaluation, findings, and recommendations.

I further declare that the analyses, visualizations, machine learning experiments, and interpretations presented in this report have been prepared for academic and educational purposes. Wherever external datasets, references, libraries, or other resources have been used, they have been appropriately acknowledged in the report.

I understand that the project is intended to demonstrate my practical understanding of data science and artificial intelligence concepts, and I take responsibility for the accuracy and authenticity of the work presented in this report.

## ACKNOWLEDGEMENT

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I am deeply thankful to **Dr. Shaheena Salam**, my project supervisor, for her guidance, support, and valuable suggestions throughout the development of this project. Her direction helped me understand the practical aspects of applying data science techniques to a real-world credit risk problem.

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Finally, I would like to thank everyone who directly or indirectly supported me during the preparation of this project report. Their encouragement and assistance were valuable throughout the project.

## ABSTRACT

This project develops a data driven credit risk classification workflow for predicting whether a credit card client will default on the following month's payment. The provided dataset contains 30,000 observations and 25 columns covering demographic attributes, credit limits, six months of repayment status, statement balances, previous payments and a binary default outcome. Data-quality checks found no missing values and no duplicate records. Rare education and marriage codes were consolidated for analysis, while legitimate monetary outliers were retained. Exploratory and statistical analysis showed that repayment behaviour was more strongly associated with default than the demographic variables, with the maximum historical payment-delay status and the number of months with delayed payments providing particularly strong signals. Logistic regression, random forest and XGBoost were compared. The final XGBoost model achieved 75.32% accuracy, 45.81% precision, 63.45% recall, 53.21% F1-score, 0.782 ROC-AUC and 0.556 PR-AUC on the held-out test set. Five-fold cross-validation produced a mean ROC-AUC of 0.787 (SD 0.006). The results demonstrate useful predictive signal in historical repayment behaviour, while also showing that the model is not a causal explanation or a substitute for governance and human review.

**Keywords:** [1]credit risk, [2]credit card default, [3]classification, [4]EDAX, [5]GBoost, [6]predictive analytics

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# 1. INTRODUCTION

## 1.1 Background

Credit risk assessment estimates the likelihood that a customer will fail to meet a credit obligation. For card issuers and lenders, early identification of elevated risk can support portfolio monitoring, credit-limit management, collections planning and customer-level review. Machine learning can complement conventional rules by learning patterns from historical repayment behaviour and financial attributes. The selected task is binary classification because the supplied target, `default.payment.next.month`, records whether a client defaulted in the following month. The IDRA guide positions credit-risk/default prediction as a classification project and requires an end-to-end workflow from data understanding through modelling and interpretation.

## 1.2 Context of the Problem

The data represent credit-card clients in Taiwan and include repayment status, statement balances, payment amounts and demographic characteristics. The study asks whether information available before the target month can distinguish subsequent default from non-default. The population should not be interpreted as representative of all current borrowers.

## 1.3 Scope of the Study

The study covers data quality, preprocessing, EDA, statistical analysis, feature engineering, supervised classification and evaluation. It does not establish causation, estimate institution-wide loss, or claim that the model is ready for production deployment.

## 2. PROBLEM STATEMENT AND MOTIVATION

Credit card lending has become an important component of the modern financial system, allowing individuals to access short-term credit and make purchases without immediate payment. However, the extension of credit also exposes financial institutions to the risk of customer default, where a borrower fails to meet the required payment obligations. Identifying customers who are more likely to default is therefore an important data-driven problem for credit providers. Traditional credit assessment approaches often rely on fixed rules, historical thresholds, and individual financial indicators. While these methods can be useful, they may not fully capture complex relationships among repayment behaviour, credit limits, billing amounts, previous payment history, and demographic characteristics.

The objective of this project is to develop a machine learning-based predictive system for assessing the likelihood of credit card payment default using historical borrower information. The dataset contains 30,000 customer records with variables describing credit limit, age, gender, education, marital status, repayment status over previous months, bill amounts, and previous payment amounts. The target variable indicates whether a customer defaulted on the following month's payment. The project aims to systematically investigate the characteristics associated with default, identify important predictive variables, and develop a classification model capable of distinguishing between customers who are likely and unlikely to default. The analysis includes data cleaning, preprocessing, exploratory data analysis, statistical testing, feature engineering, model development, and evaluation using metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Particular attention is given to repayment history and payment-delay patterns because these variables can provide valuable information about a borrower's previous credit behaviour. The final model is intended as a decision support tool, rather than a replacement for professional credit assessment, and its predictions should be interpreted alongside other relevant financial and contextual information.

The motivation for this project comes from the increasing importance of data-driven decision-making in financial risk management. Credit card providers process large volumes of customer and transaction information, creating an opportunity to use machine learning to identify patterns that may not be easily recognized through manual analysis. An effective predictive model can help financial institutions understand the characteristics of customers associated with higher default risk and support more informed approaches to credit monitoring, customer segmentation, and risk management. Early identification of potentially risky accounts may also allow institutions to take appropriate preventive measures, such as reviewing credit exposure, improving customer communication, or providing suitable repayment assistance.

From a data science perspective, credit default prediction provides a practical problem for applying a complete machine learning workflow to a real-world dataset. The problem involves several important challenges, including class imbalance, categorical and numerical variables, different scales of measurement, historical behavioural information, and the need to balance different classification objectives. For example, a model that focuses only on overall accuracy may fail to identify a significant proportion of customers who eventually default. Therefore, examining precision, recall, F1-score, ROC-AUC, and PR-AUC provides a more comprehensive understanding of model performance. The project also provides an opportunity to compare different machine learning approaches, investigate the statistical relationships between borrower characteristics and default behaviour, and interpret the features that contribute most strongly to predictions. Ultimately, the project is motivated by the goal of demonstrating how structured data analysis and machine learning can transform historical credit information into actionable insights for credit risk assessment, while recognizing the limitations and uncertainty inherent in predictive models.

### 3. OBJECTIVES

The following are the objectives that should be fulfilled by this project.

- **To quantify the factors associated with credit card default** by analyzing demographic, credit-limit, billing, payment, and repayment-history variables and identifying statistically meaningful relationships with the default outcome.
- **To determine the most influential borrower characteristics and repayment patterns** associated with the likelihood of future credit card default through exploratory, statistical, and feature-importance analysis.
- **To develop a predictive classification model** capable of distinguishing between borrowers who are likely to default and those who are unlikely to default, using historical borrower and payment information.
- **To achieve and assess predictive performance using multiple measurable metrics**, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC, rather than relying solely on overall accuracy.
- **To evaluate the reliability and generalization of the developed model** by comparing training and testing performance and applying cross-validation to measure the stability of its predictive performance.
- **To generate actionable credit risk insights** by translating the model results and statistical findings into practical recommendations for identifying and monitoring borrowers with elevated predicted default risk.

## 4. RESEARCH QUESTIONS

The given below are the research questions that guides this analysis :

- What are the main structural and data-quality characteristics?
- What borrower characteristics and financial behaviours are most strongly associated with credit card payment default?
- How does repayment behaviour, particularly the frequency and severity of previous payment delays, influence the likelihood of future default?
- Are there significant differences in default rates across borrower groups based on credit limit, education, marital status, and other demographic characteristics?
- Which variables provide the greatest predictive value for distinguishing borrowers who default from those who do not?
- How accurately can machine learning models predict future credit card default, and which evaluation metrics best reflect their ability to identify potential defaulters?
- Does the selected model demonstrate stable performance on unseen data, and what insights can be derived from its predictions for credit-risk assessment?

## 5. DATASET DESCRIPTION AND DATA UNDERSTANDING

### 5.1 Data Source

The CSV file dataset used was provided in the LMS, under the name “(P5 Dataset) - Credit Risk Assessment: Predicting Loan Default Using Customer and Loan Characteristics” in the study material provided for the interns of IDRA Training Cohort 2026.

### 5.2 Dataset Size and Structure

The dataset contains 30,000 observations (rows) and 25 variables (columns). The dataset represents individual credit card customers and contains demographic information, credit-limit information, historical bill amounts, previous payment amounts, and repayment-status information. The structure of the dataset is appropriate for a supervised binary classification problem because each customer record is associated with a known subsequent payment outcome.

The primary objective of the analysis is to predict whether a customer will default on their credit card payment in the following month. Therefore, the variable `default.payment.next.month` is selected as the target variable. It has two possible outcomes: 0, representing no default, and 1, representing default. This variable is the outcome that the machine learning models are designed to predict, while the remaining relevant customer and financial variables are used as predictors.

| Attribute              | Roles                                                                      |
|------------------------|----------------------------------------------------------------------------|
| Dataset                | UCI Credit Card Default Dataset                                            |
| File                   | P_5_UCI_Credit_Card.csv                                                    |
| Number of observations | 30,000                                                                     |
| Number of variables    | 25                                                                         |
| Prediction task        | Binary Classification                                                      |
| Target variable        | <code>default.payment.next.month</code>                                    |
| Target = 0             | Customer did not default                                                   |
| Target = 1             | Customer defaulted                                                         |
| Customer identifier    | ID                                                                         |
| Main predictor groups  | Demographic, credit limit, repayment status, bill amounts, payment amounts |

**Table 1. Summary of the Dataset**

### 5.3 Data Dictionary

| Variable                   | Meaning                                               | Role          |
|----------------------------|-------------------------------------------------------|---------------|
| ID                         | Unique identifier assigned to each customer           | Identifier    |
| LIMIT_BAL                  | Amount of credit granted to the customer              | Feature       |
| SEX                        | Gender code of the customer                           | Feature       |
| EDUCATION                  | Education-level code                                  | Feature       |
| MARRIAGE                   | Marital-status code                                   | Feature       |
| AGE                        | Age of the customer                                   | Feature       |
| PAY_0                      | Repayment status for the most recent month            | Feature       |
| PAY_2–PAY_6                | Repayment status for previous months                  | Features      |
| BILL_AMT1–BILL_AMT6        | Monthly bill statement amounts                        | Features      |
| PAY_AMT1–PAY_AMT6          | Previous monthly payment amounts                      | Features      |
| default.payment.next.month | Whether the customer defaulted in the following month | <b>Target</b> |

Table 2. Data Dictionary

## 6. DATA CLEANING

### 6.1 Missing Values

No missing values were found in the raw data, so no observations were dropped for missingness. The modelling pipeline nevertheless includes imputers as a defensive measure for future data

### 6.2 Duplicate Records

No exact duplicate rows were found, so the row count remained 30,000.

### 6.3 Data Types and Inconsistent Values

All raw columns loaded as numeric fields. Education includes rare codes 0, 5 and 6 in addition to 1–4, and marriage includes a rare 0. For analysis, 0/5/6 in education were consolidated into the other/unknown category and marriage 0 into other. Original fields remain available.

### 6.4 Outlier Handling

Many monetary observations fall beyond the  $1.5 \times \text{IQR}$  rule. These were retained because unusually large credit limits, bills or payments can be genuine financial observations. Removing them without domain evidence could discard important high-value clients.

### 6.5 Before vs After Cleaning

| Data Quality Check | Before Cleaning                        | Cleaning / Treatment Applied                                       | After Cleaning                   |
|--------------------|----------------------------------------|--------------------------------------------------------------------|----------------------------------|
| Number of rows     | 30,000                                 | No records removed                                                 | 30,000                           |
| Number of columns  | 25                                     | Original variables retained                                        | 25                               |
| Missing values     | 0                                      | No imputation required for the raw dataset                         | 0                                |
| Duplicate records  | 0                                      | No duplicate records removed                                       | 0                                |
| Data types         | Numeric variables stored appropriately | Data types reviewed and validated                                  | Valid numeric structure          |
| Education codes    | Codes 0, 1, 2, 3, 4, 5, 6              | Rare/unknown codes 0, 5 and 6 grouped with category 4 for analysis | Standardized analytical grouping |
| Marriage codes     | Codes 0, 1, 2, 3                       | Code 0 grouped with category 3 for analysis                        | Standardized analytical grouping |

|                        |                                                    |                                                                          |                                         |
|------------------------|----------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------|
| Outliers               | Extreme values present in some financial variables | Retained because they may represent genuine customer financial behaviour | Retained with appropriate preprocessing |
| Target variable        | default.payment.next.month                         | No modification to target                                                | 0 = No Default; 1 = Default             |
| Overall dataset status | Raw dataset                                        | Quality checks and justified transformations applied                     | Ready for analysis and modelling        |

Table 3. Data Quality Before and After Cleaning

## 7. DATA PREPROCESSING

The cleaned dataset was prepared for exploratory analysis and machine learning by separating the prediction target from the explanatory variables, creating behaviour-based features, encoding categorical variables, and scaling numerical variables where required. The preprocessing strategy was designed to preserve information while preventing **data leakage** between the training and testing datasets.

### 7.1 Categorical Encoding

Categorical variables and repayment-status codes are one-hot encoded. This avoids treating nominal codes as continuous quantities and allows the model to learn category-specific effects.

### 7.2 Feature Scaling and Transformation

Continuous features are median-imputed and standardised within the pipeline. Transformations are fitted only on the training partition and then applied to test data.

### 7.3 Train-Test Split

An 80 and 20 stratified split with `random_state=42` was used. Stratification preserves the target proportion and the pipeline prevents leakage from test data into preprocessing.

### 7.4 Final Feature Set

After data cleaning and feature engineering, the following variables were used for model training.

#### **Input features (predictors):**

##### **1. Demographic / categorical features**

- SEX
- EDUCATION\_GROUP
- MARRIAGE\_GROUP
- PAY\_0
- PAY\_2
- PAY\_3
- PAY\_4
- PAY\_5
- PAY\_6

## 2. Credit and financial features

- LIMIT\_BAL
- AGE
- BILL\_AMT1 – BILL\_AMT6
- PAY\_AMT1 – PAY\_AMT6

## 3. Engineered behavioural features

- MAX\_PAY\_DELAY – maximum repayment delay across the six months
- AVG\_PAY\_STATUS – average repayment status
- NUM\_DELAYED\_MONTHS – number of months with delayed payment
- TOTAL\_BILL\_6M – total bill amount over six months
- TOTAL\_PAYMENT\_6M – total payments over six months
- AVG\_BILL\_6M – average bill amount
- AVG\_PAYMENT\_6M – average payment amount
- PAYMENT\_TO\_BILL\_RATIO – payments relative to total billed amount
- AVG\_UTILIZATION – average credit utilization
- LAST\_MONTH\_UTILIZATION – credit utilization in the latest month

### Target variable:

- default.payment.next.month
  - **0:** No default in the following month
  - **1:** Default in the following month

The ID column was **excluded from modelling** because it is an identifier and does not provide meaningful predictive information.

In total, 35 original/engineered feature columns were supplied to the preprocessing and modelling pipeline, with categorical variables subsequently one-hot encoded.

## 8. EXPLORATORY DATA ANALYSIS

### 8.1 Univariate Analysis

```
# Univariate Analysis

# Target variable
sns.countplot(x='default.payment.next.month', data=df)
plt.title('Distribution of Credit Card Default')
plt.show()

# Numerical features
numerical_features = [
    'LIMIT_BAL', 'AGE',
    'BILL_AMT1', 'BILL_AMT2', 'BILL_AMT3',
    'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6',
    'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3',
    'PAY_AMT4', 'PAY_AMT5', 'PAY_AMT6'
]

df[numerical_features].hist(figsize=(16, 12), bins=30)
plt.tight_layout()
plt.show()

# Categorical features
categorical_features = [
    'SEX', 'EDUCATION', 'MARRIAGE',
    'PAY_0', 'PAY_2', 'PAY_3',
    'PAY_4', 'PAY_5', 'PAY_6'
]

for feature in categorical_features:
    sns.countplot(x=feature, data=df)
    plt.title(f'Distribution of {feature}')
    plt.show()
```

The default rate is 22.12%, so the target is moderately imbalanced. This motivates class-aware evaluation.

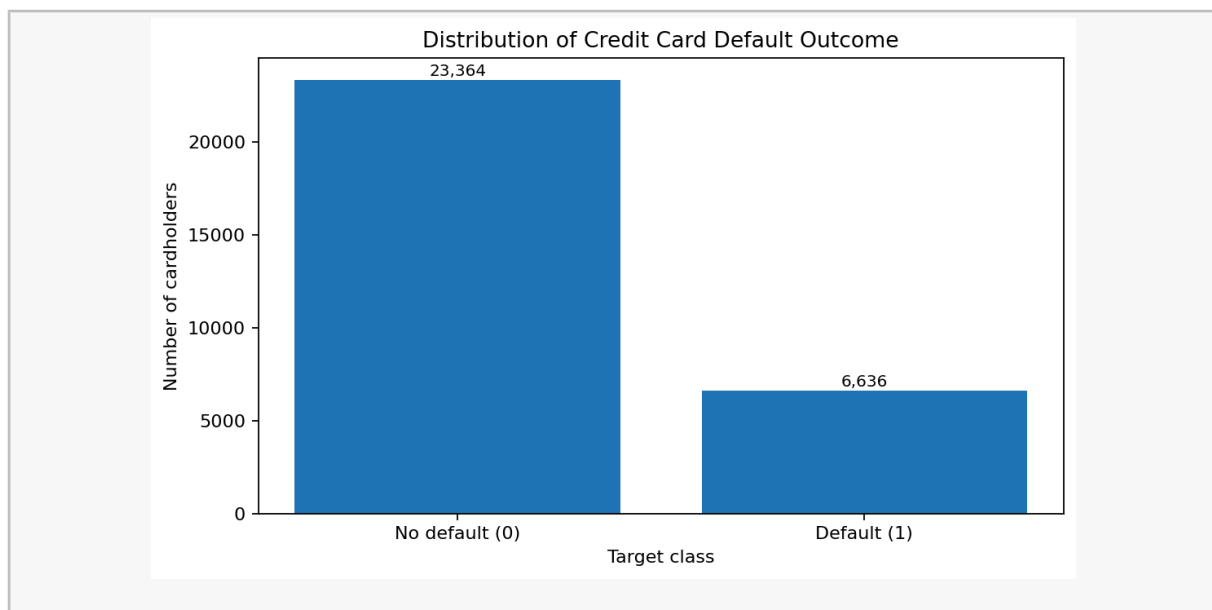


Figure 1. Distribution of the next-month default target.

The univariate analysis shows that the target variable is imbalanced, with 77.88% non-default and 22.12% default customers. Age is approximately right-skewed, while credit limits, bill amounts, and payment amounts are strongly right-skewed with some extreme values. Repayment status is mainly concentrated around on-time or low-delay categories. Some categorical groups are more frequent than others. The observed extreme financial values were retained as they may represent genuine customer behaviour.

## 8.2 Categorical Analysis

```
# Categorical Analysis

categorical_features = [
    'SEX', 'EDUCATION_GROUP', 'MARRIAGE_GROUP',
    'PAY_0', 'PAY_2', 'PAY_3',
    'PAY_4', 'PAY_5', 'PAY_6'
]

for feature in categorical_features:
    pd.crosstab(
        df[feature],
        df['default.payment.next.month'],
        normalize='index'
    ).plot(kind='bar', figsize=(7, 4))

    plt.title(f'{feature} vs Default')
    plt.xlabel(feature)
    plt.ylabel('Proportion')
    plt.legend(['No Default', 'Default'])
    plt.tight_layout()
    plt.show()
```

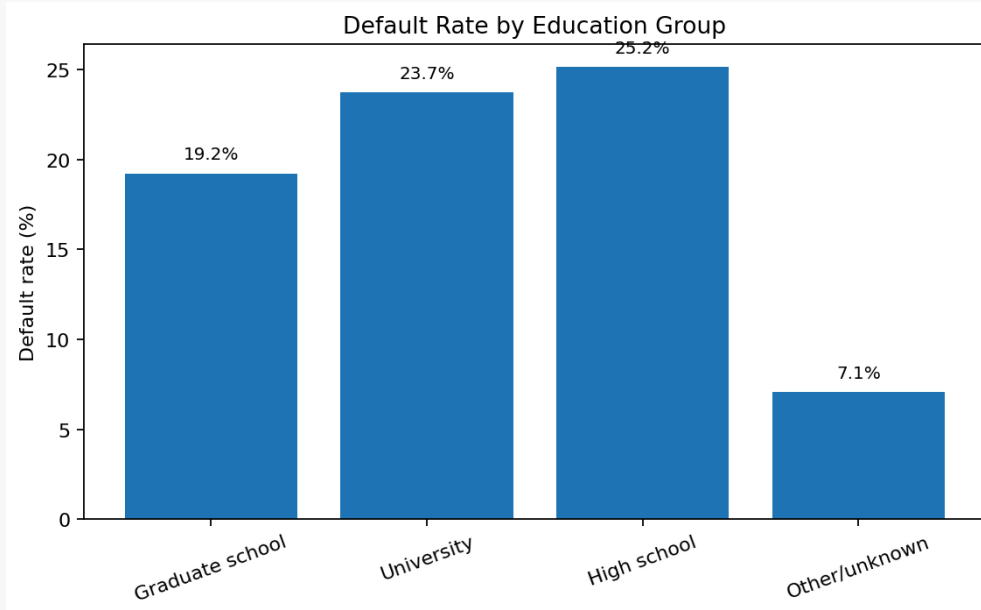


Figure 2. Default rate by education group.

Education-group default rates are approximately 19.2% for graduate school, 23.7% for university, 25.2% for high school and 7.1% for the small other/unknown group. The small final group should be interpreted cautiously.

The categories show differences in **default rates and customer characteristics**. Customers with higher repayment delays generally have a higher default rate, while differences across gender, education, and marriage categories are smaller. These differences matter because they help identify which customer groups and repayment behaviours are more associated with credit card default and can improve the model's ability to distinguish between default and non-default customers.

### 8.3 Bivariate / Relationship Analysis

```
# Bivariate / Relationship Analysis

# Correlation heatmap
plt.figure(figsize=(14, 10))
sns.heatmap(df.corr(numeric_only=True), cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.show()

# Key numerical features vs target
for feature in ['LIMIT_BAL', 'AGE', 'BILL_AMT1', 'PAY_AMT1']:
    sns.boxplot(x='default.payment.next.month', y=feature, data=df)
    plt.title(f'{feature} vs Default')
    plt.show()

# Repayment status vs default
sns.barplot(
    x='PAY_0',
    y='default.payment.next.month',
    data=df
)
plt.title('Repayment Status vs Default Rate')
plt.ylabel('Default Rate')
plt.show()
```

Default rates rise sharply with the severity of historical payment delay. This relationship is predictive rather than causal.

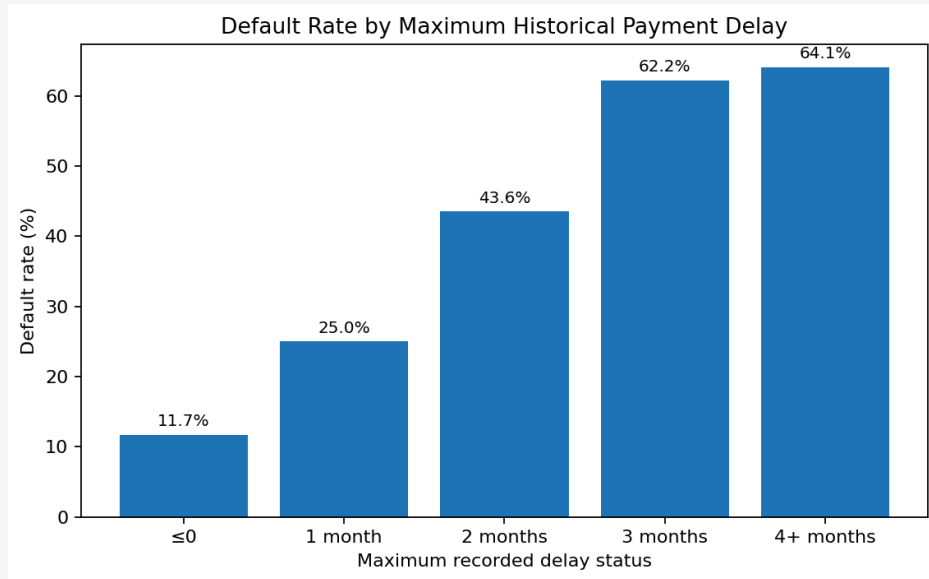


Figure 3. Default rate by maximum historical payment delay.

The median credit limit is NT\$150,000 for non-default observations and NT\$90,000 for default observations, but the distributions overlap substantially

The analysis shows a positive relationship between repayment delays and default, with NUM\_DELAYED\_MONTHS showing the strongest association ( $r \approx 0.40$ ), followed by MAX\_PAY\_DELAY ( $r \approx 0.33$ ). AVG\_PAY\_STATUS also has a moderate positive relationship ( $r \approx 0.28$ ). In contrast, LIMIT\_BAL has a weak negative relationship ( $r \approx -0.15$ ), while payment amounts show weak negative relationships. Practically, customers with more frequent or severe repayment delays tend to have a higher likelihood of default. These are associations only and do not establish causation.

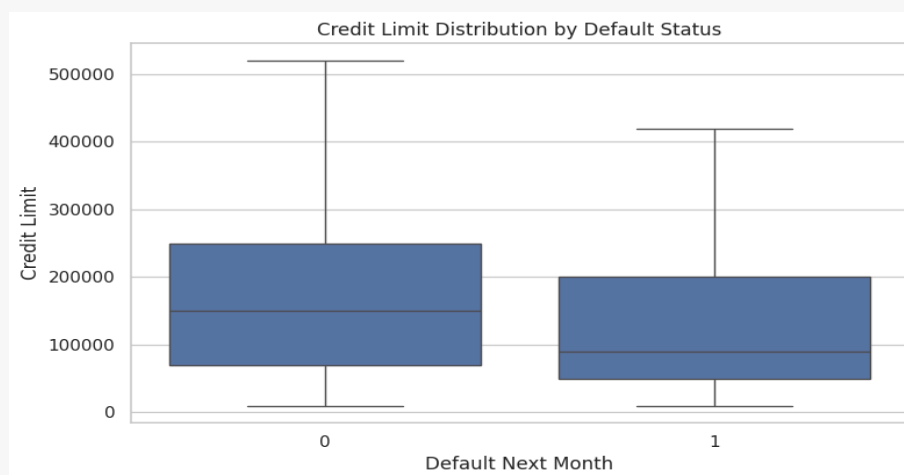


Figure 4. Credit-limit distribution by default outcome; extreme points are hidden for readability.

## 8.4 Correlation Analysis

```
# Correlation Analysis

corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(14, 10))
sns.heatmap(
    corr_matrix,
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    center=0
)
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()
```

The strongest point-biserial correlations are NUM\_DELAYED\_MONTHS (about 0.398), MAX\_PAY\_DELAY (about 0.331), AVG\_PAY\_STATUS (about 0.282), AVG\_UTILIZATION (about 0.115) and LIMIT\_BAL (about -0.154). Correlation does not establish causation.

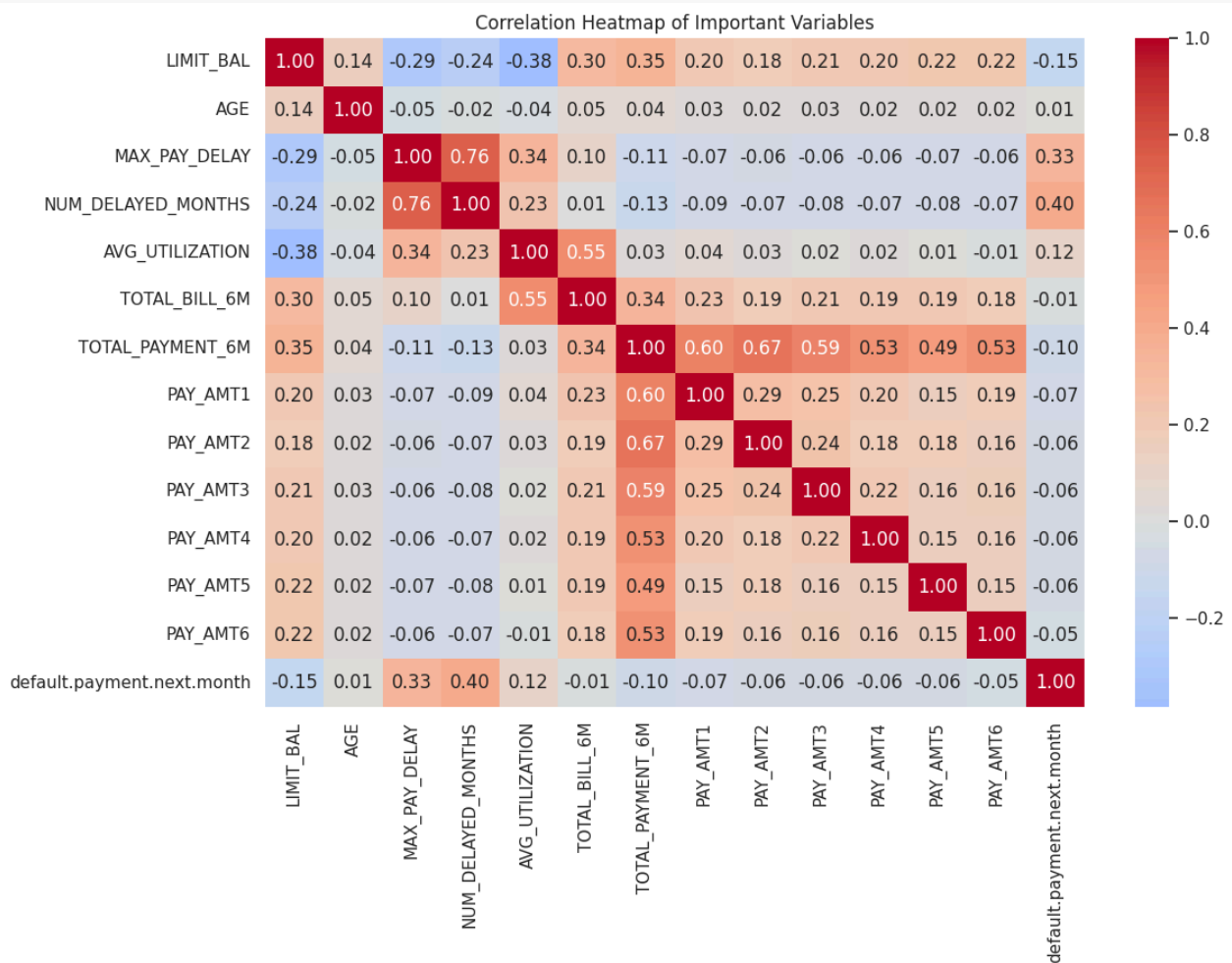


Figure 5. Correlation Heatmap of Numerical Variables

## 8.5 Other Figures

```
# Other Figures

# Default rate by education group
sns.barplot(
    x='EDUCATION_GROUP',
    y='default.payment.next.month',
    data=df
)
plt.title('Default Rate by Education Group')
plt.ylabel('Default Rate')
plt.show()

# Default rate by maximum payment delay
sns.barplot(
    x='MAX_PAY_DELAY',
    y='default.payment.next.month',
    data=df
)
plt.title('Default Rate by Maximum Payment Delay')
plt.ylabel('Default Rate')
plt.show()

# Credit limit distribution by default status
sns.boxplot(
    x='default.payment.next.month',
    y='LIMIT_BAL',
    data=df
)
plt.title('Credit Limit by Default Status')
plt.show()
```

The figures show that default rates vary across education groups, while customers with higher maximum repayment delays tend to have higher default rates. The credit-limit boxplot shows that defaulting customers generally have lower credit limits than non-defaulting customers. These patterns indicate that repayment behaviour and credit-related characteristics contain useful information for predicting default, although the figures show associations rather than causation.

## 9. STATISTICAL ANALYSIS

### 9.1 Descriptive Statistics

Non-default clients have a mean credit limit of about NT\$178,100, compared with NT\$130,110 for defaulting clients. Mean and median differences in payment-delay variables are larger and align with the EDA findings.

| Variable           | No Default Mean | No Default Median | Default Mean | Default Median |
|--------------------|-----------------|-------------------|--------------|----------------|
| LIMIT_BAL          | 178,099.73      | 150,000.00        | 130,109.66   | 90,000.00      |
| AGE                | 35.42           | 34.00             | 35.73        | 34.00          |
| MAX_PAY_DELAY      | 0.20            | 0.00              | 1.27         | 2.00           |
| NUM_DELAYED_MONTHS | 0.50            | 0.00              | 2.00         | 1.00           |
| AVG_UTILIZATION    | 0.35            | 0.24              | 0.45         | 0.46           |
| TOTAL_PAYMENT_6M   | 34,969.42       | 16,524.00         | 19,969.29    | 9,674.50       |
| TOTAL_BILL_6M      | 272,428.91      | 128,660.50        | 260,822.96   | 118,688.00     |

Table 4.Descriptive statistics of key variables by target class.

The descriptive statistics show clear differences between the two target groups. Customers who defaulted generally had lower credit limits and made lower total payments than non-defaulting customers. They also showed more frequent and severe repayment delays, as indicated by higher MAX\_PAY\_DELAY and NUM\_DELAYED\_MONTHS. Defaulting customers had higher average credit utilization, suggesting greater use of their available credit. In contrast, age was very similar between the two groups. Overall, repayment behaviour, credit utilization, and payment patterns show more noticeable differences than age.

## 9.2 Association Tests

| Variable        | Chi-square | df |
|-----------------|------------|----|
| SEX             | 47.71      | 1  |
| EDUCATION_GROUP | 160.41     | 3  |
| MARRIAGE_GROUP  | 28.13      | 3  |

Table 5. Chi-square association tests for categorical variables and default.

Chi-square tests show statistically detectable associations between default and sex, education group and marriage group (all  $p < 0.001$ ), but effect sizes are small: Cramer's V is about 0.040, 0.073 and 0.031 respectively. With 30,000 observations, statistical significance does not necessarily imply large practical effects.

## 10. FEATURE ENGINEERING

Engineered variables summarise six months of information available before the target month. They capture severity, frequency, exposure and payment activity without using future information.

| Engineered Feature     | Logic                               | Purpose / Expected Benefit                             |
|------------------------|-------------------------------------|--------------------------------------------------------|
| MAX_PAY_DELAY          | Maximum value of PAY_0–PAY_6        | Captures the most severe repayment delay.              |
| AVG_PAY_STATUS         | Mean of repayment-status variables  | Represents overall repayment behaviour.                |
| NUM_DELAYED_MONTHS     | Count of months with payment delay  | Measures the frequency of delayed payments.            |
| TOTAL_BILL_6M          | Sum of BILL_AMT1–BILL_AMT6          | Represents total billing burden over six months.       |
| TOTAL_PAYMENT_6M       | Sum of PAY_AMT1–PAY_AMT6            | Captures overall payment activity.                     |
| AVG_BILL_6M            | Mean of six monthly bill amounts    | Represents typical monthly billing level.              |
| AVG_PAYMENT_6M         | Mean of six monthly payment amounts | Represents typical monthly payment behaviour.          |
| PAYMENT_TO_BILL_RATIO  | Total payments ÷ total bills        | Indicates the proportion of billed amounts being paid. |
| AVG_UTILIZATION        | Average bill amount ÷ credit limit  | Measures overall credit utilization.                   |
| LAST_MONTH_UTILIZATION | Latest bill amount ÷ credit limit   | Captures recent credit utilization.                    |

**Table 6. Engineered Features**

These features transform the monthly billing, payment, and repayment-history variables into measures of **payment behaviour, repayment delays, financial burden, and credit utilization**, providing additional information for predicting future default.

## 11. MODEL DEVELOPMENT

### 11.1 Problem Formulation

This project is a classification task because the target variable, `default.payment.next.month`, has two discrete outcomes: 0 (no default) and 1 (default).

The input features consist of demographic, credit, billing, payment, repayment-status, and engineered behavioural variables. The target variable is `default.payment.next.month`. This formulation is appropriate because the objective is to predict whether a customer will default on their credit card payment in the following month, rather than predict a continuous numerical value.

### 11.2 Model Selection

XGBoost (Extreme Gradient Boosting) was selected as the final model because it is well suited to structured tabular data and can capture complex, non-linear relationships between customer characteristics, repayment behaviour, and default risk. It also works effectively with a mixture of numerical and categorical-derived features and can model interactions between variables.

Two alternative models were also considered: Logistic Regression and Random Forest. Logistic Regression provides a simple and interpretable baseline, while Random Forest can capture non-linear patterns through multiple decision trees. These models were compared with XGBoost using classification performance metrics before selecting the final model.

### 11.3 Training the Model

Main parameters present in this model are `n_estimators` controls the number of trees, `max_depth` controls tree complexity, and `learning_rate` controls the contribution of each tree. `subsample` and `colsample_bytree` reduce overfitting by using subsets of observations and features. `min_child_weight` and `reg_lambda` provide additional regularization. `scale_pos_weight` helps address the class imbalance in the target variable.

```
from xgboost import XGBClassifier

xgb_model = XGBClassifier(
    n_estimators=400,
    max_depth=3,
    learning_rate=0.04,
    min_child_weight=5,
    subsample=0.85,
    colsample_bytree=0.85,
    reg_lambda=2,
    scale_pos_weight=negative / positive,
    eval_metric='logloss',
    random_state=42
)

xgb_model.fit(X_train, y_train)

y_pred = xgb_model.predict(X_test)
y_prob = xgb_model.predict_proba(X_test)[:, 1]
```

## 12. MODEL EVALUATION

### 12.1 Classification Evaluation

The held-out test set contains 6,000 observations. Because default is the minority class, accuracy is interpreted alongside precision, recall, F1, ROC-AUC and PR-AUC.

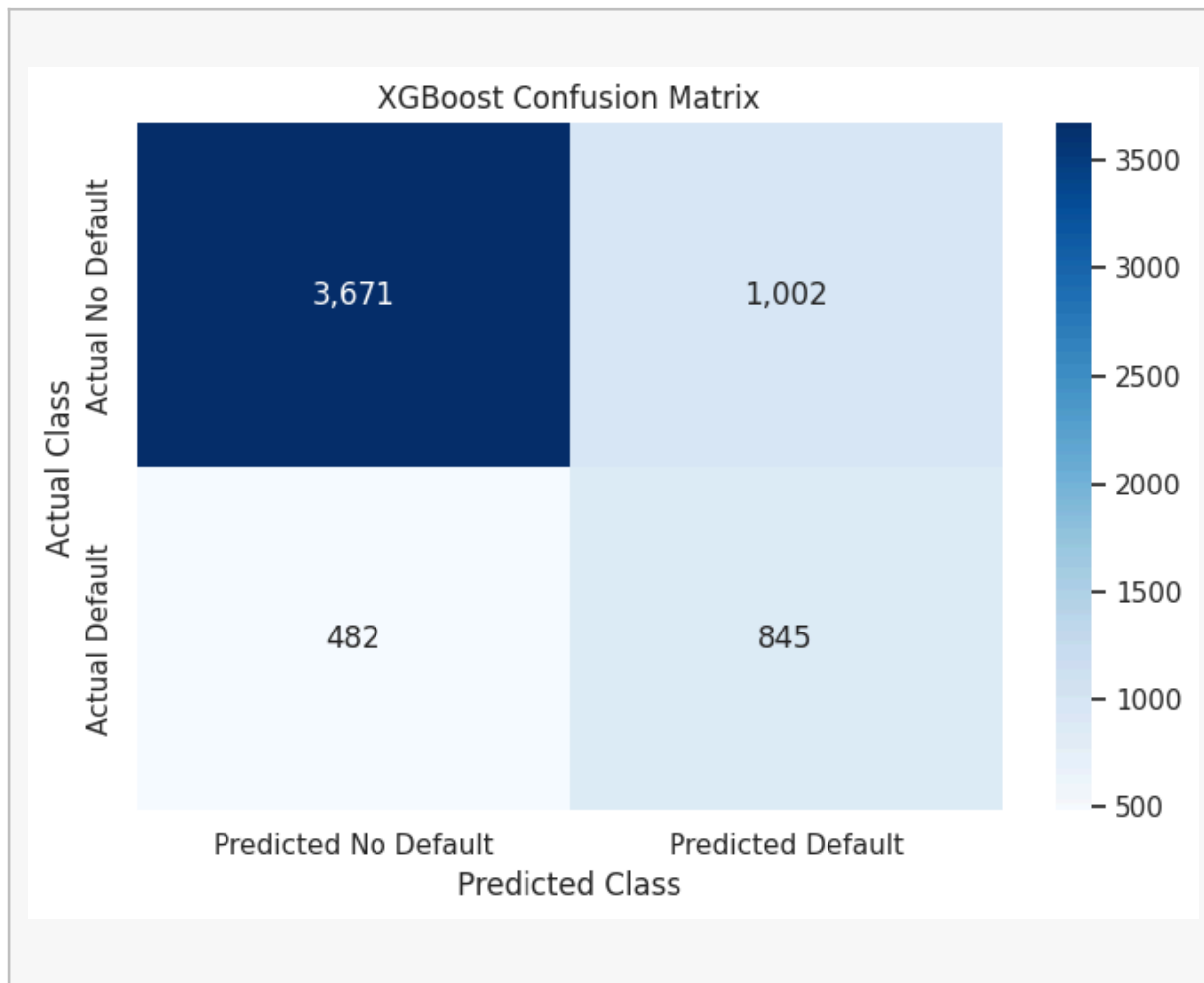


Figure 6. Confusion Matrix

| Model               | Accuracy | Precision | Recall | F1    | ROC-AUC | PR-AUC |
|---------------------|----------|-----------|--------|-------|---------|--------|
| Logistic Regression | 0.758    | 0.463     | 0.592  | 0.519 | 0.761   | 0.521  |
| Random Forest       | 0.800    | 0.550     | 0.540  | 0.545 | 0.778   | 0.554  |
| XGBoost             | 0.753    | 0.458     | 0.635  | 0.532 | 0.782   | 0.556  |

Table 7. Classification Model Performance on unseen data

## 12.2 Overfitting and Underfitting

The final model has training ROC-AUC 0.821 versus test ROC-AUC 0.782. Training PR-AUC is 0.608 versus test PR-AUC 0.556. The gap indicates some overfitting, while the stable five-fold cross-validation result supports useful generalisation within the same dataset distribution.

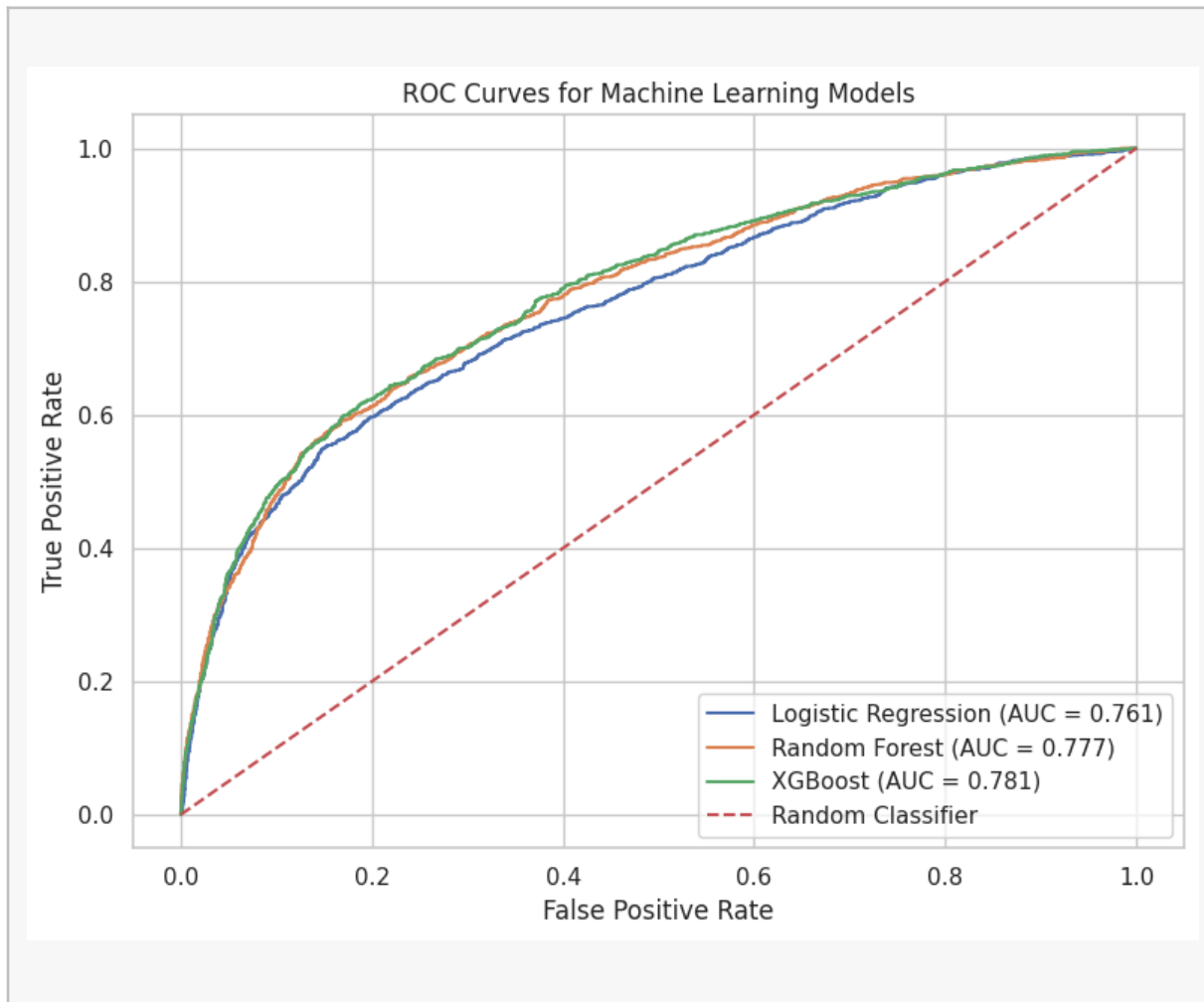


Figure 7. ROC curves for the candidate models on the held-out test set.

### 13. RESULTS AND FINDINGS

1. Finding 1: Default prevalence is 22.12%, so class-aware metrics are necessary.
2. Finding 2: NUM\_DELAYED\_MONTHS and MAX\_PAY\_DELAY are the strongest observed signals in both statistical analysis and model importance.
3. Finding 3: Default rates increase strongly with more severe historical payment delays.
4. Finding 4: Credit limit and utilisation provide additional but weaker information; no single monetary variable is sufficient.
5. Finding 5: The final model reaches ROC-AUC 0.782 and PR-AUC 0.556 on unseen data, showing useful but imperfect predictive discrimination.

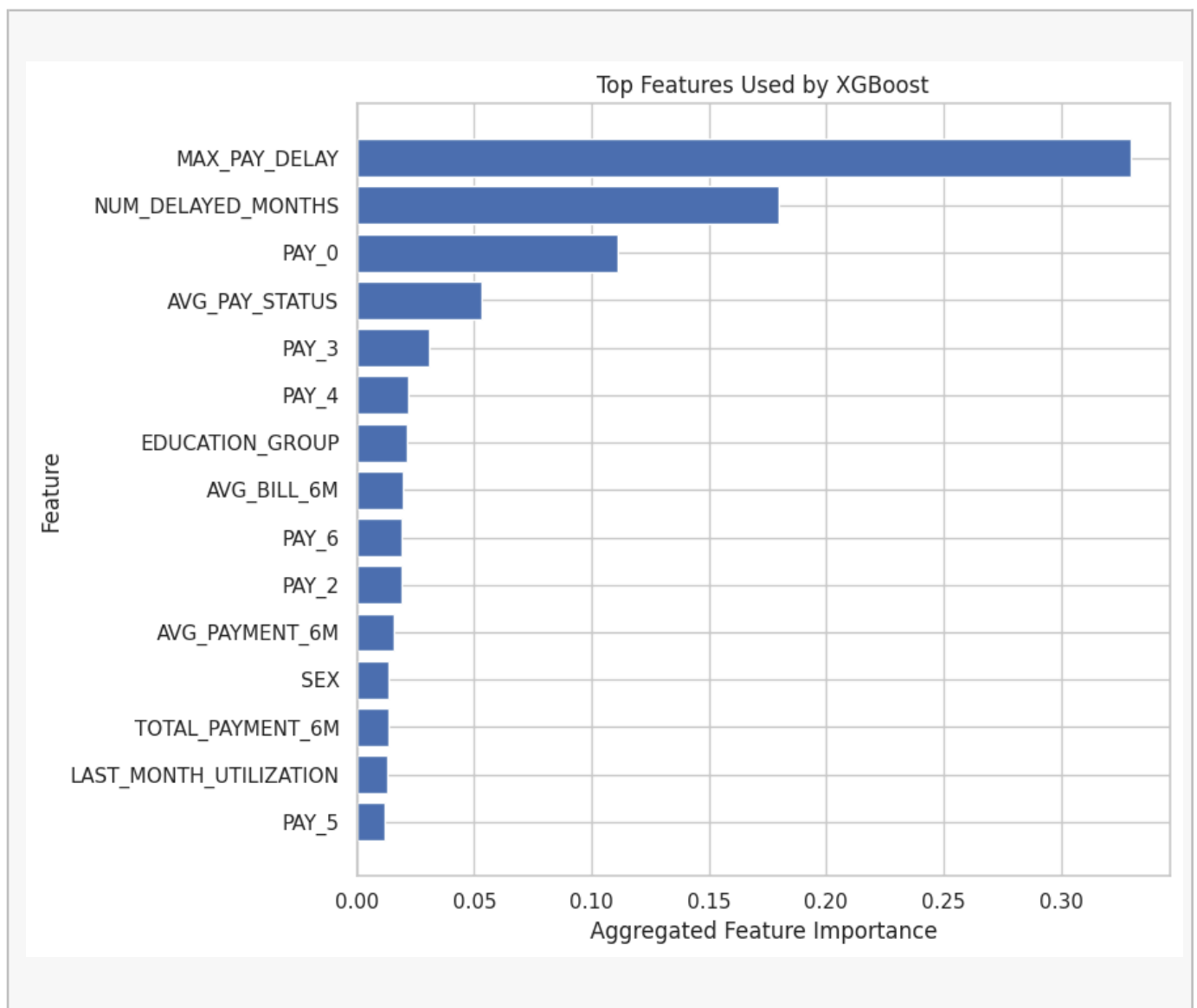


Figure 8. Aggregated XGBoost feature importance. Importance is predictive contribution, not causality.

## 14. DISCUSSION

EDA, statistical analysis and model interpretation produce a consistent pattern: recent repayment behaviour is the most informative part of the dataset. Demographic variables are statistically associated with the target, but their effect sizes are small relative to the behavioural variables. The engineered summaries capture both severity and persistence of repayment problems.

The model is predictive rather than causal. Feature importance identifies what the model uses, not what would cause an individual to default. Current deployment would require validation on contemporary, institution-specific data because borrower behaviour, policies and economic conditions can change.

## 15. LIMITATIONS

The limitations observed in the model are:

1. The dataset represents Taiwanese credit-card clients and should not automatically be generalised to other populations or products.
2. The historical period may not represent current borrower behaviour or economic conditions.
3. Important variables such as income, employment, debt-to-income and broader credit-bureau information are absent.
4. The final model shows some train-test performance gap, so overfitting remains a consideration.
5. The analysis establishes associations and predictive patterns, not causal explanations.
6. Demographic attributes may raise fairness and governance concerns and should be reviewed before any high-impact deployment.

## 16. CONCLUSION

This project completed an end-to-end credit-risk data-science workflow. The 30,000-row dataset was inspected, cleaned, transformed and analysed; meaningful visualisations and statistical summaries were produced; and three classification models were compared. Historical repayment behaviour emerged as the strongest predictive signal. The tuned XGBoost model achieved 0.782 ROC-AUC and 0.556 PR-AUC on the unseen test set. The model therefore demonstrates useful predictive signal while remaining a research prototype that requires external validation, calibration and governance before operational use.

## 17. RECOMMENDATIONS AND FUTURE WORK

### 17.1 Recommendations

Few recommendations that can improve this work are:

1. Prioritise repayment-history variables in future credit-risk monitoring because they show the strongest predictive signal.
2. Choose operating thresholds using the actual costs of false negatives and false positives rather than automatically using 0.50.
3. Monitor model performance over time and validate against newer data.
4. Use the model as one component of a broader risk-assessment process rather than as an automatic decision-maker.
5. Conduct fairness and governance reviews before operational use of demographic variables.

### 17.2 Future Work

1. Perform out-of-time validation on a later dataset.
2. Add income, employment, debt-to-income, credit-bureau and macroeconomic variables where
3. legitimately available.
4. Evaluate probability calibration and threshold optimization.
5. Use SHAP or similar explanation methods to complement feature importance.
6. Perform subgroup fairness, stability and drift analysis before deployment.

## 18. REFERENCES

- [1] Yeh, I.-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473–2480. DOI: 10.1016/j.eswa.2007.12.020.
- [2] India Data Research Academy. (2026). *Data Science & AI Capstone Project Guide 2026*.
- [3] India Data Research Academy. (2026). *Data Science & AI , LMS, “(P5 Dataset) - Credit Risk Assessment: Predicting Loan Default Using Customer and Loan Characteristics”, dataset*.
- [4] India Data Research Academy. (2026). *IDRA Capstone Project Template Sample*.
- [5] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

## APPENDIX A. CODE

The PDF contains only focused excerpts, following the IDRA guidance.

### A.1 Data Loading and Inspection

```
df = pd.read_csv("P_5_UCI_Credit_Card.csv")
print(df.shape)
print(df.isna().sum().sum())
print(df.duplicated().sum())
```

### A.2 Data Cleaning

```
df["EDUCATION_GROUP"] = df["EDUCATION"].replace({0:4,5:4,6:4})
df["MARRIAGE_GROUP"] = df["MARRIAGE"].replace({0:3})
```

### A.3 Preprocessing

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, stratify=y,
random_state=42)
preprocessor = ColumnTransformer([('num', StandardScaler(), numerical_features), ('cat',
OneHotEncoder(handle_unknown='ignore'), categorical_features)])
X_train = preprocessor.fit_transform(X_train)
X_test = preprocessor.transform(X_test)
```

### A.4 EDA and Visualization

```
sns.countplot(x='default.payment.next.month', data=df) plt.show()
sns.histplot(df['LIMIT_BAL'], kde=True) plt.show()
sns.boxplot(x='default.payment.next.month', y='LIMIT_BAL', data=df) plt.show()
sns.barplot(x='PAY_0', y='default.payment.next.month', data=df) plt.show()
sns.heatmap(df.corr(numeric_only=True), cmap='coolwarm') plt.show()
```

### A.5 Feature Engineering

```
df["MAX_PAY_DELAY"] = df[pay_cols].max(axis=1)
df["NUM_DELAYED_MONTHS"] = (df[pay_cols] > 0).sum(axis=1)
df["TOTAL_BILL_6M"] = df[bill_cols].sum(axis=1)
df["TOTAL_PAYMENT_6M"] = df[payamt_cols].sum(axis=1)
```

### A.6 Model Training and Evaluation

```
pipeline.fit(X_train, y_train)
prob = pipeline.predict_proba(X_test)[: , 1]
pred = (prob >= 0.50).astype(int)
print(confusion_matrix(y_test, pred))
print(roc_auc_score(y_test, prob))
```

## APPENDIX B. ADDITIONAL FIGURES AND TABLES

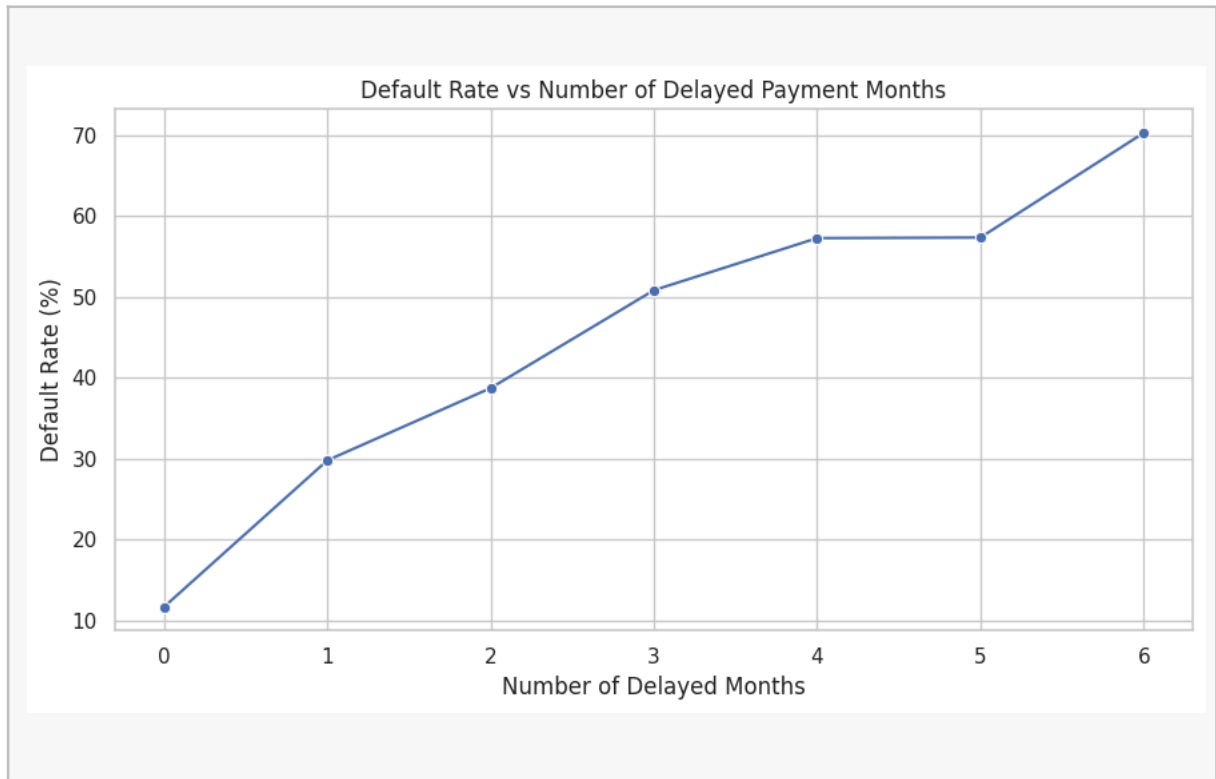


Figure 9. Default Rate vs. Number of Delayed Payment Months

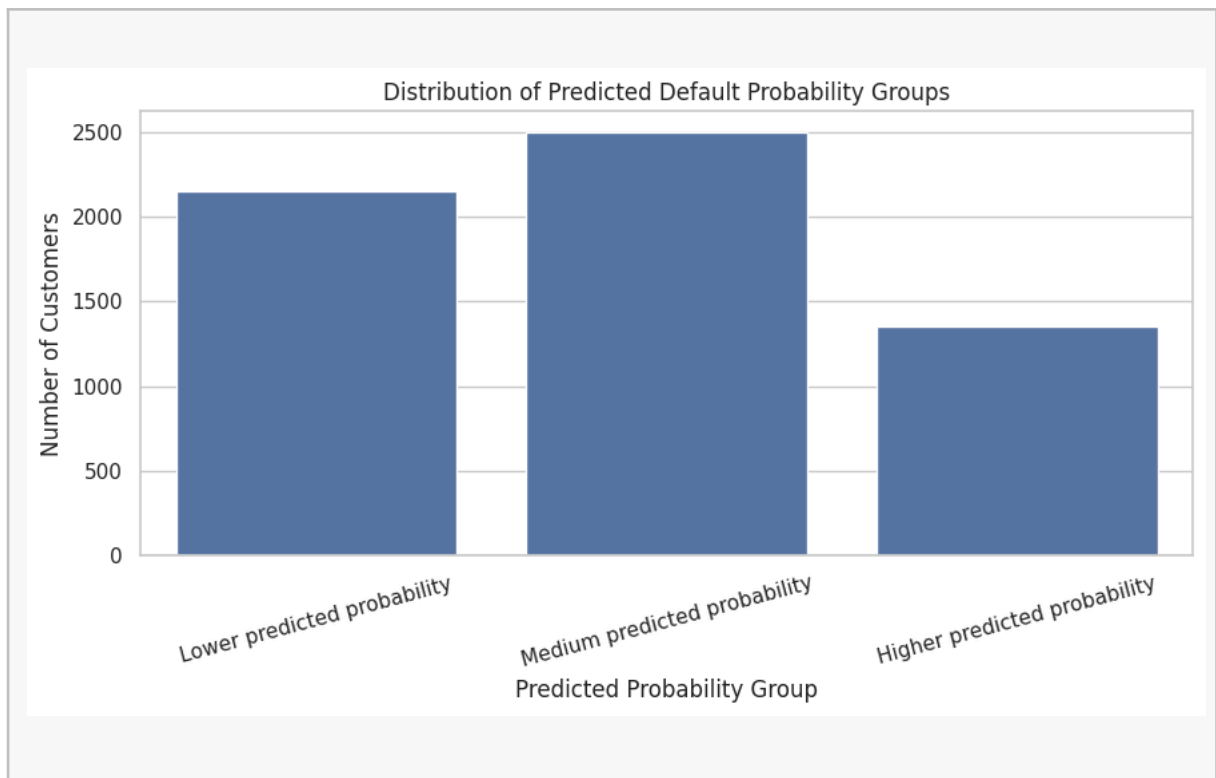


Figure 10. Distribution of Predicted Default Probability Groups

| default.payment.next.month | proportion |
|----------------------------|------------|
| 0                          | 77.88      |
| 1                          | 22.12      |

Table 8. Target distribution (%)

|    | Variable  | Outliers | Lower Limit | Upper Limit |
|----|-----------|----------|-------------|-------------|
| 0  | LIMIT_BAL | 167      | -235000.000 | 525000.000  |
| 1  | BILL_AMT1 | 2400     | -91739.625  | 162389.375  |
| 2  | BILL_AMT2 | 2395     | -88547.500  | 155538.500  |
| 3  | BILL_AMT3 | 2469     | -83581.500  | 146412.500  |
| 4  | BILL_AMT4 | 2622     | -75942.125  | 132774.875  |
| 5  | BILL_AMT5 | 2725     | -70878.250  | 122831.750  |
| 6  | BILL_AMT6 | 2693     | -70657.375  | 121111.625  |
| 7  | PAY_AMT1  | 2745     | -5009.000   | 11015.000   |
| 8  | PAY_AMT2  | 2714     | -5417.500   | 11250.500   |
| 9  | PAY_AMT3  | 2598     | -5782.500   | 10677.500   |
| 10 | PAY_AMT4  | 2994     | -5279.875   | 9589.125    |
| 11 | PAY_AMT5  | 2945     | -5416.000   | 9700.000    |
| 12 | PAY_AMT6  | 2958     | -5705.625   | 9823.375    |

Table 9.IQR outlier analysis

|    | Feature                | Correlation | p_value       |
|----|------------------------|-------------|---------------|
| 16 | NUM_DELAYED_MONTHS     | 0.398394    | 0.000000e+00  |
| 14 | MAX_PAY_DELAY          | 0.331036    | 0.000000e+00  |
| 15 | AVG_PAY_STATUS         | 0.281955    | 0.000000e+00  |
| 0  | LIMIT_BAL              | -0.153520   | 1.302244e-157 |
| 22 | AVG_UTILIZATION        | 0.115481    | 1.400908e-89  |
| 18 | TOTAL_PAYMENT_6M       | -0.102354   | 1.118695e-70  |
| 20 | AVG_PAYMENT_6M         | -0.102354   | 1.118695e-70  |
| 23 | LAST_MONTH_UTILIZATION | 0.086166    | 1.521053e-50  |
| 8  | PAY_AMT1               | -0.072929   | 1.146488e-36  |
| 9  | PAY_AMT2               | -0.058579   | 3.166657e-24  |
| 11 | PAY_AMT4               | -0.056827   | 6.830942e-23  |
| 10 | PAY_AMT3               | -0.056250   | 1.841770e-22  |
| 12 | PAY_AMT5               | -0.055124   | 1.241345e-21  |
| 13 | PAY_AMT6               | -0.053183   | 3.033589e-20  |
| 2  | BILL_AMT1              | -0.019644   | 6.673295e-04  |

Table 10.Point-biserial correlations

## Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| No Default   | 0.88      | 0.79   | 0.83     | 4673    |
| Default      | 0.46      | 0.64   | 0.53     | 1327    |
| accuracy     |           |        | 0.75     | 6000    |
| macro avg    | 0.67      | 0.71   | 0.68     | 6000    |
| weighted avg | 0.79      | 0.75   | 0.77     | 6000    |

Table 11.Classification Report

| Risk_Group                   | count |
|------------------------------|-------|
| Lower predicted probability  | 2152  |
| Medium predicted probability | 2496  |
| Higher predicted probability | 1352  |

Table 12.Predicted probability groups